Specification of Feature Isolation

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1 Basic Information and Results

Feature isolation is based on the following information:

- A specification of a software product line including the names of its independent features, the feature categories to be considered, the maximum rank of interaction between any kind of and- and or-features, and optionally explicitly dependent features plus available systems or missing systems all of which resulting in a set with the identifiers of available systems. This specification is denoted by *SPL*. An identifier (*ID* for short) is a natural number with a lower bound of 1.
- For each feature there exists a description that specifies in which systems it occurs and in which it does not plus an associated unique feature *ID*. Such a feature description is called *FD*, a feature *ID* is called *FID*.
- Each system has a description that specifies the features it consists of plus an associated unique system *ID*. Such a system description is called *SD*, a system *ID* is called *SID*.

Evaluation. To describe the isolation of a feature f_i based on its *FID* some operations are introduced.

- Given is a system description *SD*.
- *SD* can be flattened by replacing all each of its elements by the corresponding set of *FID*s and applying either set intersection, $\bigcap$, or set union, $\bigcup$, to them.
- Flattening *SD* with $\bigcap$ gives $\widehat{SD}$.
- Flattening *SD* with $\bigcup$ gives $\breve{SD}$.

The isolation of a feature f_i specified by its *FID* involves the following steps:

1. I is the set with *SID*s of all systems that include FID_i.
2. U is the set with *SID*s of all systems that do *not* include FID_i.
3. $\widehat{I} \setminus \breve{U}$ gives the result of feature isolation which is denoted by $\overline{D}$. $\overline{D}$ must not be equal to $\varnothing$ as this would indicate a fundamental error during the preceding steps. $\overline{D}$ can include exactly one element which then must be the FID_i. If $\overline{D}$ includes more than one element, one of them must be FID_i.

Minimal set differences Given are the following equations and the definition:

1. $D = I \backslash U$
2. $\overline{D} = \hat{I} \backslash \check{U}$
3. $I_S \subseteq I$
4. $U_S \subseteq U$
5. $D_S \subset D \coloneqq I_S \subset I \vee U_S \subset U$
6. $minimal(D) \coloneqq \forall_{D_S} \in \{\mathcal{P}(D) \backslash D\} : \overline{D_S} \neq \overline{D}$

From 6 follows that any D that evaluates to $\overline{D}$ if there is no proper subset of D which also gives $\overline{D}$ is *minimal*.

In addition, there exists at least one subset of D which gives $\overline{D}$, namely D. There can be more than one proper subset of D which also gives $\overline{D}$. The set that includes all different subsets of D that evaluate to $\overline{D}$, is called M. $|M| \geqslant 1$.

The following examples illustrate the meaning of *minimal* in a set difference (the elements are *FID*)s. Given are the systems
$SD_1 = \{1, 2, 3\}$,
$SD_2 = \{2, 3, 4\}$,
$SD_3 = \{3, 4, 5\}$,
$SD_4 = \{4, 5, 6\}$,
$SD_5 = \{2, 7\}$ and
$SD_6 = \{1, 8\}$.

With respect to FID_2, the following set intersections I are minimal as they all isolate FID_2.
$I_1 = SD_1 \cap SD_2$
$I_2 = SD_1 \cap SD_5$
$I_3 = SD_2 \cap SD_5$

$I_4 = SD_1 \cap SD_2 \cap SD_5$ is not minimal because I_1, I_2 and I_3 successfully isolate FID_2 but contain less operands. It is important to note that only evaluating I can give the desired result. Furthermore, I cannot be omitted.

With respect to FID_2 the set difference $D_1 = SD_1 \backslash SD_3 \cup SD_6$ is minimal as it isolates FID_2.

$D_2 = SD_1 \cap SD_2 \backslash SD_3$ is not minimal, as I already isolates FID_2. $D_3 = SD_1 \backslash SD_3 \cup SD_4 \cup SD_6$ is also not minimal, as $SD_3 \cup SD_6$ in U already removes all unwanted elements of I. It has to be emphasized, that —different from I— U can be omitted.

2 Functions, Datatypes and Variables

- The datatypes `feature_ids_t` and `system_ids_t` represent the set types for *FID*s and *SID*s.
- The member variable `spl_specification_t::m_available_systems` which is of type `system_ids_t` contains the *SID*s of the actually available systems.
- `difference_t` is the type that represents a single set difference. It defines the data members `I` and `U`, both of type `system_ids_t`.They represent the operands I and U of a set difference that isolates a feature f_i.
- `minimal_differences_t` represents M. It defines the data members `features` which is of type `feature_ids_t` and represents D